JCUIScaler 1.0 – Static Code Analysis

Rev L – document closed for version 1.0 (macOS, move to GitHub)

Compilation: FPC 3.2.2 / LCL 3.0 without warnings, with and without LCLGTK2; packages jcuiscaler_rt and jcuiscaler_dsgn (lazbuild). Runtime tests: GTK2 (Xvfb), 96 and 120 DPI — no drift; the component is pixel-identical to the manual integration. Windows (Lazarus 4.8): demo, RestoreInitialScale and IDE installation of the component confirmed by the author. Delphi 13.2 (Win64): compilation and basic operation confirmed by the author, including third-party controls (TMS UI Pack PageControl, RxDBGrid). macOS 27 (Lazarus 4.8, Cocoa): demo verified by the author.

1. Scope of the analysis

The analysis covers the modules JCUIScaler.pas, JCUIScaler.capture.inc, JCUIScaler.controls.inc and JCUIScaler.find.inc (the .inc files were renamed in RC1.2; findings F-01..F-11 use the names valid when they were reported). Rev E documents the state of the code after the cleanup session of 2026-06-01, carried out on the basis of the Rev D findings. Rev G covers a code review for compatibility with Delphi VCL and a test compilation for FPC/LCL with the GTK2 widgetset. The new findings F-12..F-17 are described in section 7a; all were resolved in RC1.2.

Rev H covers the first runtime tests of the library outside Windows: the demo project built for GTK2 and run in a virtual X server at 96 and 120 DPI (resize cycles with pixel-exact screenshot comparison, all tabs, second form). Findings F-18..F-22 are described in section 7b. None of them required a change to the library code.

Rev I covers the new RC2 modules: JCUIScalerComponent.pas (the TJCScaler component), design/JCUIScalerReg.pas (IDE registration) and the packages packages/lazarus/jcuiscaler_rt.lpk and jcuiscaler_dsgn.lpk, as well as the license headers and license files. The core code has not changed (changes in comments only). Findings F-23..F-27 are described in section 7c; F-22 is closed.

Rev K covers the verification with Delphi 13.2 (Win64). Findings F-28..F-29 are described in section 7d; both are closed. Behaviour under the LCL is unchanged (A/B test: 12/12 screenshots identical).

2. Architecture

The library core is based on TJCUIScaler, a structure of type object. The library does not use the class model as the basis of its operation. The scaler state is held inside the object; integration takes place through ScalerInit, ScalerOnShow and ScalerOnResize. The object lives on the stack — no explicit Create/Free is required.

3. Data model

The code keeps separate reference records for DBGrid, StringGrid, generic controls, ListView, TreeView, StatusBar, GroupBox, Splitter and BorderSpacing. The data is held in dynamic arrays built once during CaptureOriginals and is not modified during scaling.

4. CaptureOriginals

Iterative DFS with its own TWinControl stack (INITIAL_STACK_SIZE = 256, grows dynamically). Safety limit MAX_CONTROL_COUNT = 8192. The stack is released when the procedure ends. Records are initialised with Default(T) — correct practice for records containing dynamic arrays.

5. Scaling algorithm

ScaleForm computes overallScaleX and overallScaleY relative to FormOriginalWidth/Height. Scaling is performed only relative to the data from CaptureOriginals. Minimum form sizes (MIN_FORM_WIDTH/HEIGHT = 200) protect against unstable scaling. All pixel computations use the uniform LocalRound method (Floor(V + 0.5) since RC1.2, previously Trunc(V + 0.5) — see F-15) instead of the built-in Round — this eliminates the asymmetry of banker's rounding.

6. Reference data lookup

Find.inc provides six Find*OrigIndex functions — one for each special component type: FindDBGridOrigIndex, FindStrGridOrigIndex, FindCtrlOrigIndex, FindListViewOrigIndex, FindStatusBarOrigIndex, FindTreeViewOrigIndex. All use a linear search by pointer. ApplyControlExceptions uses these functions uniformly for all types.

7. Rev D findings — status after cleanup

Status legend: CLOSED = fixed in the code; DECISION = closed by a deliberate design decision (no code change); DEFERRED = planned, requires further work.

ID	Severity	Status	File / location	Description	Resolution
F-01	HIGH	DECISION	JCUIScaler.pas RestoreInitialScale	RestoreInitialScale uses a brute-force proportional restore via ScaleBackX/Y instead of restoring from the reference model.	Design decision: the brute-force approach is intentional. CtrlOrigs does not cover every control — restoring from the model would skip elements outside it. A code comment explaining this decision was added. Restoring Font.Size from OrigFontSize was added (no effect in 1.0 — font scaling is not planned for version 1.0, see section 9).
F-02	MEDIUM	CLOSED	JCUIScaler.pas ScaleControl	SavedBevel and SavedBorder declared unconditionally but assigned only in the TPanel block — risk of a compiler warning.	Variables initialised to 0 at declaration. RC1.2: initialisation by assignment at the start of the procedure (Delphi compatibility, see F-16).

F-03	MEDIUM	DECISION	JCUIScaler.Controls.inc ApplyControlExceptions	TGroupBox is registered in GroupBoxes, but ApplyControlExceptions has no TGroupBox section — a dead record in the model.	GroupBoxes registration intentionally kept: OrigFontSize reserved for the future. Font scaling is not planned for version 1.0 (see section 9) — the TGroupBox section in ApplyControlExceptions will be added when font scaling is implemented. A comment was added in Capture.inc.
F-04	MEDIUM	CLOSED	JCUIScaler.Controls.inc ApplyControlExceptions	TListView, TStatusBar, TTreeView used local find loops instead of dedicated Find* functions.	FindListViewOrigIndex, FindStatusBarOrigIndex, FindTreeViewOrigIndex added to Find.inc. ApplyControlExceptions uses them uniformly.
F-05	MEDIUM	DEFERRED	JCUIScaler.Controls.inc DBGrid / StringGrid	ClientRect after SetBounds may be stale on some widgetsets before the LCL layout is refreshed.	Deliberately deferred: the current version is stable. The change requires extensive multi-platform testing. The risk of breaking working code outweighs the potential gain in RC1.
F-06	LOW	CLOSED	JCUIScaler.pas / JCUIScaler.Controls.inc	Two different rounding functions: Round() in ScaleControl, a local LocalRound() in ApplyControlExceptions.	LocalRound extracted as a private method of TJCUIScaler. Used uniformly in ScaleControl and ApplyControlExceptions. The local nested function removed.
F-07	LOW	DECISION	JCUIScaler.pas GridLastWOffset	Field always equal to 0, not exposed as a property — ambiguous status.	Kept as a private field with an extended comment. Rationale: exotic Linux configurations (fractional DPI, specific GTK themes)

					may require a correction. Removing it is cheaper on paper than in practice given LCL's multi-platform nature.
F-08	LOW	CLOSED	JCUIScaler.Capture.inc	Bit $10000000 in Tag as an undocumented public contract.	Public constant JCSCALER_BORDER_TAG = $10000000 added to the interface section of JCUIScaler.pas, with a comment and a usage example. Capture.inc uses the constant instead of the literal.
F-09	LOW	CLOSED	JCUIScaler.Controls.inc TTreeView / LCLGTK2	Empty except without a comment — makes diagnosing platform problems harder.	Split into two separate try..except blocks (Indent / ItemHeight GTK2), each with a comment explaining why the exception is silently ignored. RC1.2: ItemHeight replaced by DefaultItemHeight (see F-14).
F-10	INFO	CLOSED	JCUIScaler.pas ScalerOnShow	The comment "// For future use" was misleading about the implementation state.	Comment updated: describes the current behaviour (ResetOnShow) and reserves room for future extensions.
F-11	INFO	CLOSED	JCUIScaler.Capture.inc TTreeView	OrigItemHeight/ OrigIndent = 0 without an explanation in the specification.	Documented in Spec R1-02 section 6.1. The FS3 comment in the code kept and extended.

7a. Rev G findings (RC1.2)

Findings detected while preparing Delphi compatibility and during the GTK2 test compilation. Status legend as in section 7.

ID	Severity	Status	File / location	Description	Resolution

F-12	HIGH	CLOSED	JCUIScaler.capture.inc BorderSpacing	TBorderOrig stored the control's position and edges (Left, Top, Left+Width, Top+Height), which were then written into BorderSpacing — a control with JCSCALER_BORDER_TAG received margins equal to its own coordinates. The feature did not work as described in Spec 5.3.	Capture stores the original BorderSpacing.Left/Top/Right/Bottom. Changed behaviour for tagged controls — requires a demo test.
F-13	MEDIUM	CLOSED	JCUIScaler.controls.inc ApplyControlExceptions	The BorderSpacing section was placed at the end of the procedure — unreachable for TDBGrid, TStringGrid, TListView, TTreeView, TStatusBar and TSplitter, whose sections end with Exit.	Section moved to the start of ApplyControlExceptions, without Exit. Grids, ListView and StatusBar compute column widths after the final BorderSpacing is set.
F-14	HIGH	CLOSED	JCUIScaler.controls.inc TTreeView / LCLGTK2	Reference to the non-existent property tv.ItemHeight (LCL TTreeView provides DefaultItemHeight) — the library did not compile with the GTK2 widgetset. Not detected in Rev D–F because the {$IFDEF LCLGTK2} block is excluded when compiling for Windows.	Replaced with DefaultItemHeight; verified by compiling with LCLGTK2. No change in behaviour (OrigItemHeight = 0, FS3). Process conclusion: test compilation with all supported widgetsets (Spec R1-04 section 14).
F-15	LOW	CLOSED	JCUIScaler.pas LocalRound	Trunc(V + 0.5) rounds negative values incorrectly (e.g. -1.7 → -1 instead of -2). Negative Left/Top occur, among others, in a scrolled TScrollBox.	Floor(V + 0.5). Identical result for V >= 0 — no impact on existing test scenarios.
F-16	MEDIUM	CLOSED	All modules Delphi VCL	Constructs not portable to Delphi VCL: {$mode objfpc}; LCLIntf/LCLType units; initialised local variables in the var section (SavedBevel/SavedBorder); TControl.Font is protected in VCL; no BorderSpacing in VCL.	Three {$IFDEF FPC} blocks (compiler mode, BorderSpacing capture and apply); unused units LCLIntf, LCLType, Dialogs, Buttons removed; initialisation by assignment; access class

					TJCControlAccess. Object architecture unchanged. In Delphi JCSCALER_BORDER_TAG is ignored.
F-17	INFO	CLOSED	Spec R1-03 sections 3, 4, 6, 8, 10, 11, 12	Discrepancies between the specification and the code: exclusion of TSpeedButton (the code excludes only TToolButton); reference model built in ScalerInit (in the code: on the first ScalerOnShow); the integration example set properties before ScalerInit, which overwrites them with defaults.	Spec R1-04 corrected to match the code. Excluding TSpeedButton was a specification error — the observed problems concerned TToolButton only.

7b. Rev H findings (runtime tests)

Findings from runtime tests of the demo on GTK2 (96 and 120 DPI) and confirmations by the author on Windows. Status legend as in section 7.

ID	Severity	Status	File / location	Description	Resolution
F-18	MEDIUM	DECISION	JCUIScaler.controls.inc TListView	A native TListView on a hidden tab sheet reports a ClientWidth without its vertical scrollbar. Columns scaled at that time are wider by the scrollbar width until the next form resize; if a horizontal scrollbar is allowed, it appears once the tab is shown. Windows and GTK2 (GTK2: 451 px hidden, 435 px visible). LCL grids are not affected — measured for ssAutoBoth, ssVertical, ssBoth, 6 and 60 rows.	No code change. Recommendation ScrollBars = ssVertical (Spec section 13.1) — confirmed on Windows and GTK2. Optionally (Windows only) ScalerOnResize in TPageControl.OnChange. Rejected: OnResize handler in the scaler (violates the design), deferred call (flicker), SM_CXVSCROLL reserve (GTK2: 16 px vs 19 px).
F-19	LOW	DECISION	LCL (DPI scaling) demo project	Controls with Anchors = [] are moved by the LCL during automatic DPI scaling (DPI different	Recommendation: default Anchors = [akLeft, akTop] (Spec section 13.1, demo

				from DesignTimePPI) before the first ScalerOnShow. The reference model records the broken layout. The effect also occurs without the scaler.	README). Demo corrected (12 controls).
F-20	INFO	DECISION	LCL GTK2 ScrollBars	In the GTK2 widgetset ScrollBars = ssBoth corresponds to GTK_POLICY_ALWAYS — both scrollbars are always visible, regardless of content. This made assessing F-18 harder.	Documented (Spec section 13.1). The demo uses ssVertical.
F-21	INFO	CLOSED	Demo project	The demo contained copies of the RC1 library, a StatusBar without panels (the panel scaling code was not tested) and did not cover JCSCALER_BORDER_TAG.	Demo rebuilt: library from ../source, StatusBar with three panels, panel with JCSCALER_BORDER_TAG on the second form, names and texts unified, Lazarus only.
F-22	MEDIUM	CLOSED	JCUIScaler.pas RestoreInitialScale	RestoreInitialScale calls FForm.SetBounds before the proportional loop over CtrlOrigs. On GTK2 OnResize (and ScaleForm from the model) follows after the method ends — result pixel-identical to the first show. Hypothesis for Windows: a synchronous WM_SIZE might trigger ScaleForm inside SetBounds, and the proportional loop would then be applied to controls already restored from the model. In addition, the code comment (acting on the entire visible state) did not match the code — the loop covers only the controls in the model.	Decision: the method remains public (emergency mechanism); Spec 8.4 and the API table updated. Windows test (second demo form: open, enlarge and move, close, reopen) — layout correct; the hypothesis was not confirmed. Code comment corrected in RC2 (matches the actual behaviour).

7c. Rev I findings (TJCScaler component, license)

Findings from the review and tests of the TJCScaler component and from preparing the license files. Status legend as in section 7.

ID	Severity	Status	File / location	Description	Resolution
F-23	INFO	CLOSED	JCUIScalerComponent.pas TJCScaler	Review of the component against the critical architecture risks (Risk Analysis, section 2 and the RC1.2 rule): the integration layer must not change the core, must not register handlers on controls and must not tie the core's lifetime to the controls.	Compliant. The component hooks only form events (OnShow — chaining, OnResize — AddHandlerOnResize) and calls only the public methods of TJCUIScaler. Core code unchanged. The TJCUIScaler object lives as a field of the component owned by the form.
F-24	MEDIUM	DECISION	JCUIScalerComponent.pas Attach / OnShow	The LCL provides no handler list for OnShow, so the component replaces the form's OnShow with its own handler and calls the original one. If the application replaces OnShow at run time (including in FormCreate), the scaler is disconnected. AddHandlerFirstShow was rejected: it fires before OnShow, so the reference model would be built from a different state than with the manual integration.	Documented (Spec R1-06, sections 10 and 12.2). The public Attach method reconnects the component and takes over the new handler. Destroy restores the application's OnShow unless the form itself is being destroyed.
F-25	INFO	DECISION	JCUIScalerComponent.pas class name	The default instance name created by the IDE from the class name (TJCUIScalerComponent → JCUIScalerComponent1) was too long. The name TJCUIScaler is taken by the core object used in production.	Component class: TJCScaler (instance JCScaler1). The core name TJCUIScaler unchanged — compatibility with applications using the manual integration preserved. Icon resources:

					TJCSCALER, _150, _200.
F-26	LOW	CLOSED	JCUIScalerComponent.pas properties	In the first version of the component, SkipBevelScaling and ResetOnShow were copied to the object once in Attach; changing them in code after the form was loaded did not reach the scaler. No extension points without touching the core.	Properties linked to the object (getters/setters). Events BeforeShow, AfterShow, BeforeResize, AfterResize and virtual methods DoShow and DoResize added. Verified: event order, ResetOnShow set after Attach works, descendant class, 12/12 screenshots identical to the manual integration.
F-27	INFO	CLOSED	License files source headers	No license headers in the source files (JCUIScaler.pas referred to a non-existent licence.txt). RoadMap R2-04 stated LGPL 2.1, whereas the LCL license is GNU LGPL version 2 with the static linking exception.	License identical to the LCL: COPYING.LGPL.txt (LGPL v2, copied from Lazarus), COPYING.modifiedLGPL.txt (exception text identical to Lazarus). LICENSE, COPYRIGHT, AUTHORS. Headers in all source files with a prohibition on removing the copyright notice. Code identical when comments are ignored.

7d. Rev K findings (Delphi 13.2)

Findings from compiling and testing with Delphi 13.2 (Win64). Status legend as in section 7.

ID	Severity	Status	File / location	Description	Resolution
F-28	MEDIUM	CLOSED	JCUIScaler.capture.inc TSplitter	Delphi E2010: in VCL TSplitter descends from TGraphicControl; the test “Root is TSplitter” with Root of type TWinControl is rejected	Test changed to “TObject(Root) is TSplitter” (no IFDEF). LCL behaviour unchanged (A/B: 12/12 screenshots

				at compile time.	identical). In VCL the splitter receives the generic bounds scaling. Compliant with section 2 of the Risk Analysis.
F-29	INFO	CLOSED	Source files line endings	The library source files had LF line endings; RAD Studio reports a warning (it requires CRLF).	All Pascal files (.pas, .inc, .lpr, .lfm) converted to CRLF; .gitattributes keeps CRLF in the repository. Lazarus accepts both.

8. Strengths — state after cleanup

+ Deterministic reference model — no accumulation of scaling errors.
+ Uniform rounding function LocalRound throughout the scaling code.
+ Consistent reference model lookups through dedicated Find* functions for all types.
+ Iterative DFS instead of recursion — safe for deep structures.
+ Public contract JCSCALER_BORDER_TAG instead of a magic literal.
+ English comments — ready for publication.
+ Source-level compatibility with Delphi VCL without changing the object architecture — three {$IFDEF FPC} blocks.
+ Demo tests after cleanup: no regressions.
+ Test compilation FPC/LCL with and without LCLGTK2 without warnings (RC1.2).
+ GTK2 runtime tests at 96 and 120 DPI: no drift after resize cycles (pixel-exact screenshot comparison).
+ TJCScaler component as a pure integration layer: results pixel-identical to the manual integration, core unchanged.

9. Open areas before OPM publication

Ref	Description
F-05	ClientRect after SetBounds — examine the behaviour on additional widgetsets when test resources become available.
—	Font scaling — rolled back because of deep DPI/platform/OS version dependencies (Windows 7/10/11, GTK2, macOS). Status after 1.0: a “help wanted” (experimental) issue in the project repository.
—	TGroupBox in ApplyControlExceptions — tied to font scaling. Deferred together with it.
R2-01	Monitor change with a different DPI (LCL) and DPI-change handling in Delphi/VCL — “help wanted” issues in the project repository.

10. Conclusions

After the cleanup of 2026-06-01 the code is architecturally and implementationally consistent. All Rev D findings were resolved: six fixed in the code, four closed by a design decision with a documented rationale, one deliberately deferred (F-05). Demo tests after the changes showed no regressions.

The library is ready for further development towards OPM publication. The main open areas blocking publication are the English documentation and the OPM package. Font scaling and TGroupBox are deliberately deferred beyond version 1.0.

Rev G (RC1.2): the Delphi compatibility review and the GTK2 test compilation revealed six new findings (F-12..F-17), including two of HIGH severity: non-working BorderSpacing scaling and failure to compile for LCLGTK2. All were closed in RC1.2. The code compiles without warnings with FPC/LCL; Delphi 13 compilation needs to be confirmed.

Rev H: the first runtime tests outside Windows confirmed the deterministic behaviour of the library (no drift) on GTK2 at two DPI values. Findings F-18..F-20 concern LCL and widgetset behaviour and were resolved with configuration recommendations without code changes; a code-change variant for F-18 was rejected as contrary to the library's design. F-21 was closed by rebuilding the demo project. F-22 (RestoreInitialScale — OnResize order on Windows) remained to be tested; the decision to keep the method public is final.

Rev I (RC2): the TJCScaler component complies with the critical architecture risks and gives results identical to the manual integration. The only design limitation (OnShow chaining, F-24) is documented and has a workaround (Attach). The license is aligned with the LCL. F-22 was closed by a test on Windows. Delphi 13 compilation remains open — planned for the end of the work.

Rev J (1.0 release): review of the state before publication — no new findings. Changes only in comments (version labels unified to “1.0”) and in repository files (.gitignore); code identical to RC2 when comments are ignored. English documentation and README completed. Still open: Delphi 13 compilation (the README states that verification is in progress) and F-05 (deliberately deferred).

Rev K: compilation and basic operation with Delphi 13.2 confirmed, including third-party controls (TMS UI Pack PageControl, RxDBGrid). The only compilation error (F-28) was removed by changing the type in an is test, with no effect on the LCL; line endings unified to CRLF (F-29). Of the open areas only the deliberately deferred F-05 and font scaling beyond 1.0 remain.

Rev L (closed for version 1.0): demo verified on macOS 27 (Lazarus 4.8, Cocoa) — tested platforms: Windows, macOS and Linux/GTK2, and unofficially Delphi 13.2. The register of findings F-01..F-29 is complete for version 1.0. This document is not updated on an ongoing basis: new findings are tracked as issues in the project repository (GitHub), and the rules for contributing changes are maintained in CONTRIBUTING.md. The current document is the Technical Specification.

Analysis revision history: Rev C → Rev D (extended with F-01..F-11) → Rev E (state after the 2026-06-01 cleanup) → Rev F (font scaling deferred beyond 1.0, demo project closed) → Rev G (RC1.2: source-level Delphi compatibility, findings F-12..F-17) → Rev H (GTK2 runtime tests, findings F-18..F-22) → Rev I (RC2: TJCScaler component, license, findings F-23..F-27, F-22 closed) → Rev J (1.0 release state) → Rev K (Delphi 13.2, findings F-28..F-29) → Rev L (closed for version 1.0).